
HPL *ink* IPTV E2+ Radio

User manual

A guide to the program for Android TV. The Polish version of this document is a separate file.

Platform

Android TV / Google TV

Edition

v2.13 31.08.2026

Sources

M3U, Xstream, Enigma2, radio

Control

TV remote + browser panel

Marks used here: **TV** means an action on the television. **PANEL** means an action in a browser on a computer or a phone, at the address shown on the TV screen.

Document edition: this manual describes HPL IPTV E2+ Radio v2.13 of 31.08.2026. The activation screen is in English. The language of the rest of the program and of the panel follows the television language (Polish or English).

English user manual

1. What the program is	9. Browser panel
2. Program activation	10. Adding sources
3. First start	11. Films and series
4. Sources screen	12. Phone remote
5. Channel list	13. Remote keys
6. Player	14. Panel settings
7. Internet radio	15. Manual EPG matching
8. Guide on the TV	16. Troubleshooting

1. What the program is

HPL IPTV E2+ Radio is an Android TV app. It plays IPTV channels, Enigma2 tuners and internet radio. Playlists, EPG, films and settings are added from a computer or phone through a browser panel — the same PC that runs HPL ink Enigma2+ IPTV, or any other browser on the home network.

- M3U / M3U8 playlist from a URL.
- Xstream Codes account (server, port, user, password).
- Enigma2 tuner with OpenWebif — bouquets, stream and guide from the receiver.
- Internet radio from an M3U playlist (URL or a file from the computer).
- XMLTV guide, films and series from the playlist, favourites, history, reminders and backup.

Networks: the TV, computer and phone must be on the same local network. The panel and the activation page are not a public website — they work only at home, at the address shown on the TV.

What is new since version 1.87

New	Where	Described in
Right opens a strip of channel tiles — logo, name and the programme on air now	Player 	chapter 6
Left opens the bouquet of the channel you are watching; left / right move between bouquets	Player 	chapter 6
A red dot under the OSD clock when a reminder is waiting	Player 	chapter 6

New	Where	Described in
A heads-up about the next programme a few minutes before it starts	Player 	chapter 6
The guide imports itself once a day and yields to playback	Guide 	chapter 8
"Pick up where you left off" tiles and a list of pending reminders on the Start page	Panel 	chapter 9
Hiding bouquets you never watch	Panel 	chapter 14
Guide status: how many programmes, how far ahead it runs, and a clear button	Panel 	chapter 14

Activation, adding sources, the phone remote and the panel tabs work exactly as in version 1.87. The OSD changed as well: the channel name is red, and the logo now sits in the middle of the field to its left.

2. Program activation

On the first start the program does not open the source list. The TV shows only the **ACTIVATION REQUIRED** screen. That screen is in English. Until a valid activation code is entered, the rest of the program stays closed.

An activation code from one television will not work on another.

Easiest path — a computer

The activation code is long. Pasting it in a computer browser is easier than typing with the TV remote.

1. Address

On the TV, read **OPEN THIS ADDRESS ON A COMPUTER**, e.g.

`http://192.168.1.50:8091/activate`.

2. Request code

Open the address in a browser and select **Copy code**.

3. Send

Send the copied code to

`hplink.soft@proton.me`.

4. Activate

Paste the code you receive into **ACTIVATION CODE** and select **Activate**.

Use this exact subject line:

`HPL IPTV E2+ Radio activation`

A successful activation shows *Activated. The program is starting on the TV*. The television then opens the normal sources screen. Later starts do not show this screen again.

The same steps on the TV

If no computer is at hand, the same fields are on the TV screen:

- **DEVICE ID** — a short identifier in the form `HPL-XXXX-XXXX-XXXX`.
- **REQUEST CODE** — the long code to send.
- **Copy code** — copies the request code (when the TV has a clipboard).
- **ACTIVATION CODE** — field for the code you receive.
- **Activate** — submits the code.
- **Close program** — closes the app.

This code is not valid for this device. means the pasted code does not match this television. Check that you sent the request code from this device and that you pasted the entire code you received, with nothing cut off.

Other versions: hplspam1-jpg.github.io/HPL-ink-Software

Network during activation: the `/activate` address works only when the TV is on the local network. If the TV shows `no network connection`, connect Wi-Fi or a cable and start the program again.

3. First start

Requirements

- A television or box with Android TV / Google TV, Android 8.0 or newer.
- The same local network for the TV, computer and phone.
- For IPTV: an M3U/M3U8 playlist or Xstream Codes details from the provider.
- For Enigma2: a receiver with OpenWebif enabled on the same network.
- For radio: an M3U playlist at a URL or a `.m3u` file from the computer (6 MB maximum).
- Optional TMDb API v3 key — only for descriptions, ratings and original titles. Posters come from the playlist.

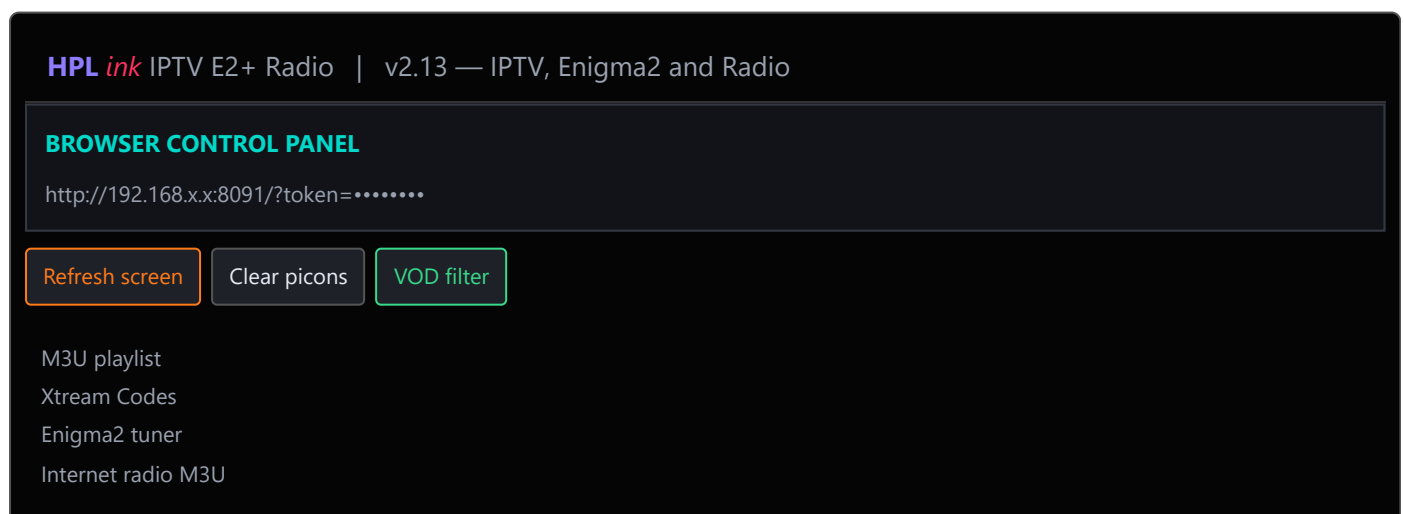
After activation

1. The TV shows the sources screen and a large panel address, e.g. `http://192.168.1.50:8091/?token=...`.
2. Copy the **entire** address, including the `token` parameter.
3. Open it in a browser on the computer.
4. In the panel, under **Settings**, add the first source.
5. Go back to the TV and open that source to load the list.

Important: an address without `token` will not open the panel. When adding a second TV, always paste the full address from its screen.

4. Sources screen TV

This is the first screen after activation. The drawing below is a UI diagram and contains no real address or token.



- **Refresh screen** — reloads the source list.
- **Clear picons** — removes cached channel logos from the TV.

- **VOD filter** — hides or shows films and series on the channel list. The same switch is in the panel.
- **OK** opens a source. Holding **OK** asks to delete it.
- **INFO** or **GREEN** tests the selected source.

Until you add a source in the panel, the list is empty. Sources are not added on the TV.

5. Channel list TV

Opening an IPTV, Xstream or Enigma2 source shows groups on the left and channels on the right. A current programme may appear next to a channel when EPG is available.

Key	Action
OK	Play the selected channel.
Search	Searches a channel name or a programme title in the guide (7 days).
GREEN	EPG (XMLTV) import for this source. An Enigma2 tuner provides the guide itself and needs no import.
BLUE	Fetches the playlist from the provider again.
YELLOW	Adds or removes the channel from favourites. The ★ Favourites group appears in the group list.
INFO or RED	Opens the guide for the selected channel. If there is no EPG, the TV suggests an XMLTV import.

Search on the TV has two modes: **Channel** (names) and **In the guide** (programme titles for 7 days). **OK** on the field opens the keyboard, **BACK** hides it.

6. Player TV

A channel opens full screen. The OSD shows the channel logo, **the channel name in red**, the programme title, broadcast times and a progress bar. It hides after a moment and returns with **OK** or **INFO** — called up by a key it stays twice as long as the one that appears by itself after a channel change.

The bottom OSD line shows **which playlist** is playing: the word **IPTV** plus the first two words of the source name — the source "TV VOD 12.2026" gives **IPTV TV VOD**. A tuner shows just **Enigma2**. Next to it sit the colour key hints and the item **hold OK — picture**.

Key	Action
CH+ / CH- or up / down	Next / previous channel.

Key	Action
OK / INFO	OSD. On a film, also pause and resume. It acts on key release , so it does not clash with the long press.
Hold OK	Picture ratio selection.
Left	Channel list (zapper) — it opens on the bouquet of the channel you are watching . On a film — back 30 s.
Right	Strip of channel tiles. On a film — forward 30 s.
RED	Return to the previously watched channel.
GREEN	Audio track selection.
YELLOW	Full description of the current programme. Up / down scrolls the text.
BLUE	Subtitle selection.
BACK	Closes an overlay; if none is open, closes the player.

In the zapper list, up / down selects a row, **OK** plays it, **left and right move to the neighbouring bouquet** — the channel you are watching keeps playing — and **BACK** closes the list. A hint with this set sits under the bouquet name.

Tile strip — the right key

On a live channel **right** slides a strip of tiles in at the bottom of the screen. Each tile carries the channel logo, its name and the programme on air there right now. The picture keeps playing underneath.

- **Right / left** moves the selection. While the strip is open, left does *not* open the channel list.
- **OK** plays the selected channel, **BACK** closes the strip.
- The strip walks the **current bouquet** and stops at its ends. For other bouquets use the list under the left key.
- Opening the strip fetches nothing from the network — logos and the guide come from the TV's own memory.

The reminder dot and the next-programme notice

- A **red dot under the clock** in the OSD means at least one reminder is waiting. It shows on every channel and goes out only when the last pending reminder has fired. What exactly is waiting is listed on the panel's Start page.
- The **next-programme notice** appears in the corner of the screen a few minutes before that programme starts. How far ahead is set in the panel (chapter 14); it can also be switched off.
- A dropped stream is retried several times. If the same folder has a backup link, the program switches to it and shows a failover notice. When there is no replacement it says so: *no backup link for this channel*.
- Film position is stored. The next opening resumes from that point.

- If the picture freezes or the channel does not respond after the retries, a message with the attempt count is shown.
- A minute of healthy picture clears the memory of failed links — after a short network overload the program has somewhere to switch again.

Picture ratio

Holding **OK** opens the **Picture ratio** window — the same layout as the audio track selection. It works on a live channel and on a film. The active mode is marked.

Mode	What it does
Stretch (as before)	The picture fills the whole screen. The default, and the behaviour known from earlier versions.
Fit — whole picture	The whole frame is visible, with bars at the top and bottom or at the sides.
Zoom — no bars	The screen is filled without distortion; the edges of the frame go past the screen.

The choice is stored, shared by all channels, and survives a channel change and a restart of the program.

16:9 material looks the same in every mode — there is nothing to fit. The difference shows on cinema films and on old 4:3 channels.

When the picture stops or the network drops

- **The picture froze while the sound keeps playing.** The program counts the frames actually drawn on the screen. When none arrives for a dozen or so seconds it shows *No picture detected*. *Restoring the picture* and rebuilds the connection on its own. Such a freeze used to pass unnoticed, because the time counter kept moving along with the sound.
- **The network dropped.** After a lost Wi-Fi connection or a router restart the screen shows *No network · waiting for a connection*. The program does not burn through its retries and does not declare the channel dead. When the network returns it shows *Network is back · resuming playback* and picks the picture up again — without leaving the channel. Radio behaves the same way, and the browser panel comes back up after the TV changes address.

7. Internet radio **TV**

A radio source opens a station list, not a TV channel list.

- **OK** — play the selected station.
- **BLUE** — refresh the playlist.
- **YELLOW** — favourites.

In the radio player: **OK** pause, up / down change station, left / right volume, **BACK** return to the list. A clock is shown, and the bitrate when the station provides it.

Radio reacts to the network like the channel player: after a lost connection it waits with *No network · waiting for a connection* and resumes the station by itself once the network is back.

8. Guide on the TV **TV**

INFO on the channel list opens the full schedule for one channel. Programmes are grouped by day. **OK** on an item can play the channel now or set a reminder. A reminder can also switch the TV at the start time.

The guide refreshes itself once a day. Never while you are watching — the job waits in a queue and starts when nobody is watching. If an import is already running and you start a channel, it stops at the nearest step and returns to the queue. A television switched off for a few days catches up at its next start.

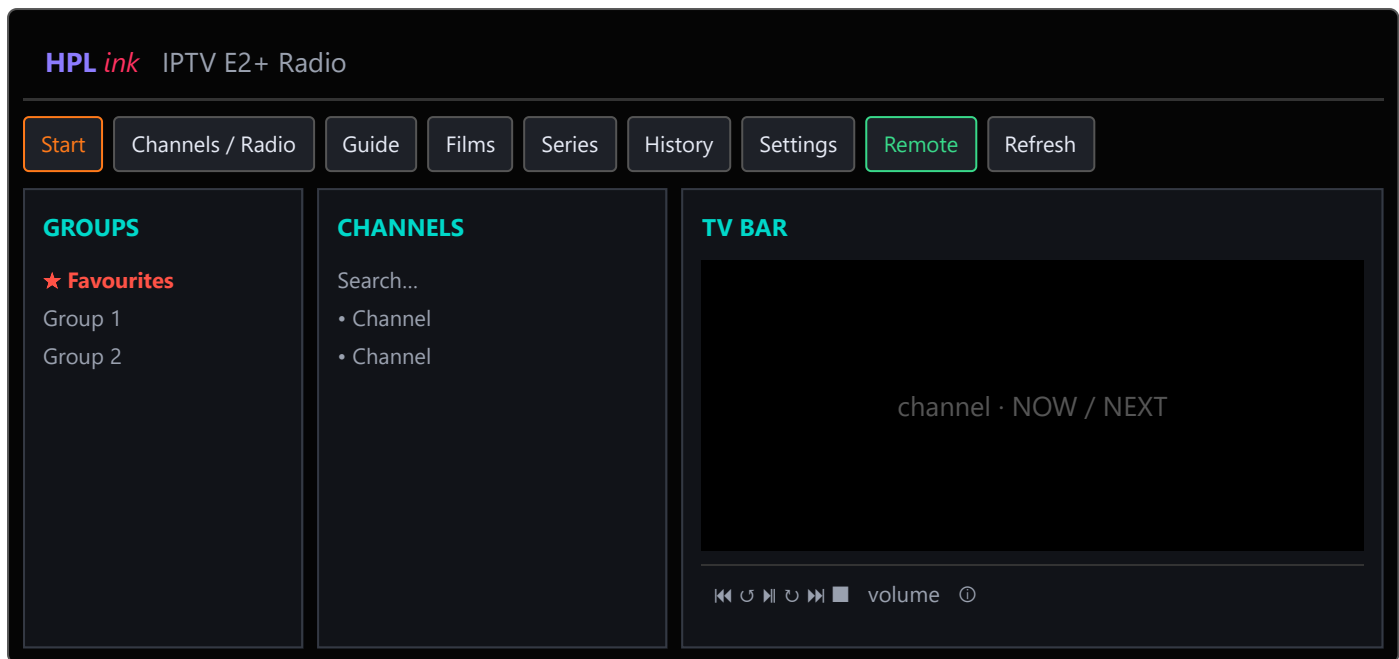
A manual XMLTV import is started with **GREEN** on the channel list or with the button in the panel. Download, name matching and storage run on the TV in the background and may take a few minutes. An Enigma2 tuner needs no import — the guide comes from OpenWebif. Radio does not use the TV guide.

An interrupted import no longer wipes the guide. New programmes go to a holding area first, and the old guide lives until the whole file has been read correctly. The swap happens in one step. A dropped network or a damaged XMLTV file therefore leaves the guide as it was — not empty or half filled. While importing, the program also clears programmes older than half a day and further away than eight days.

A channel the automatic matching cannot recognise by name can be assigned to an XMLTV entry by hand, from the panel — see chapter 15.

9. Browser panel **PANEL**

The panel is the page opened from the address on the TV screen. Its language follows the app language on the television, not the browser language.



Tabs

- **Start** — at the top, **Pick up where you left off** tiles: the last channel and the last radio station, one click plays them on the TV. Below them a list of **pending reminders** with a cross to delete one — shown only when something is waiting, which is exactly when the red dot glows in the OSD. Then: now on favourites, continue watching, series in progress, to watch and recently added.
- **Channels / Radio** — groups, list, favourites filter and an all-sources switch.
- **Guide** — EPG grid, timeline shift, programme search over 7 days.
- **Films** and **Series** — poster wall from the playlist.
- **History** — recently watched items, with a clear action.
- **Settings** — sources, TV sets, EPG, backup, TMDb, QR remote.

TV bar

The top bar shows the selected TV, the current channel, the programme and a recording mark when the Enigma2 tuner is recording. Buttons: previous / next channel, back and forward 30 s (films), pause, stop, volume, mute and ⓘ — show the panel on the TV. The seek slider appears for films only.

The **Source** list switches the playlist. **Refresh** fetches the list from the TV into the browser again. Search everywhere (Ctrl+K) covers channels, films and series. The microphone button starts voice control in browsers that support it.

Several TV sets: one panel can control several televisions. Switching TV also changes lists and the guide, because each TV has its own sources.

10. Adding sources PANEL

In **Settings** → **Sources** → **Add a source** choose the type and fill in the fields. After adding, open the source on the TV to load the list.

M3U playlist

1. Type: **M3U playlist**.
2. Name, e.g. "Home list".
3. Playlist address, e.g. `http://server/list.m3u`.
4. Optional EPG (XMLTV) address.
5. **Add**.

Xtream Codes

1. Type: **Xtream Codes**.
2. Server address, port (usually 80), user and password from the provider.
3. Optional EPG address.
4. **Add**.

Enigma2 tuner

1. Host Receiver address on the local network.	2. Port OpenWebif, usually 80. User and password if authentication is on.	3. Stream Stream port, usually 8001.	4. Add Open the source on the TV — bouquets will load.
--	--	--	--

From the panel, an Enigma2 programme can be scheduled as a recording on the tuner. The TV asks for confirmation, and the receiver must accept the timer.

Internet radio

- **URL address** — an M3U playlist on the network.
- **.m3u file from the computer** — the file is checked and stored in the app on the TV. Limit 6 MB. The file must contain valid `#EXTINF` entries.

Privacy: adding or browsing a list does not open every stream. A provider connection is made when an item is played or when a playlist / EPG / logo is fetched.

11. Films and series PANEL

The **Films** and **Series** tabs show items that the playlist marks as VOD (paths or metadata such as `/movie/` and `/series/`). Posters come from the playlist. A TMDb key in Settings adds the description, rating and original title only after an item is opened.

- Sorting: newest, oldest, playlist order, most episodes (series).
- Filters: with poster only, to watch.

- The **+** badge on a poster marks an item to watch.
- In series: folders → series → season → episode. **Next unwatched** resumes from the first unwatched episode.
- **Play on the TV** starts the item on the selected television.

The VOD filter on the TV sources screen and in panel Settings hides these items on the ordinary channel list so they do not mix with live TV.

12. Phone remote PANEL

Settings shows a QR code. Scan it with a phone on the same network — a separate, narrow remote opens in the phone browser.

- TV and source selection.
- Current channel, EPG and a film seek slider.
- Previous / next, pause, ± 30 s, stop, mute, volume, panel on the TV.
- Tabs: favourites, folders, recent, search.

The **Remote** button in the panel header opens the same window on the computer.

13. Remote keys — summary

Screen	Key	Action
Sources	OK	Open the source. Hold to delete.
Sources	INFO / GREEN	Connection test.
Channels	OK	Play.
Channels	GREEN	EPG import.
Channels	BLUE	Refresh the playlist.
Channels	YELLOW	Favourites.
Channels	INFO / RED	Channel guide.
Player	Up / down / CH $\pm$	Change channel.
Player	Hold OK	Picture ratio.
Player	Left	Channel list of the current bouquet, or -30 s in a film.
Player	Right	Tile strip, or $+30$ s in a film.
Channel list in the player	Left / right	Neighbouring bouquet.

Screen	Key	Action
Tile strip	Left / right, OK , BACK	Pick a channel, play it, close the strip.
Player	RED	Previously watched channel.
Player	GREEN	Audio track.
Player	YELLOW	Programme description.
Player	BLUE	Subtitles.
Radio	Up / down	Station.
Radio	Left / right	Volume.
Web panel	Ctrl+K	Search everywhere.

14. Panel settings PANEL

- 1. Remote on your phone.** QR code and remote page address. The phone must be on the same network.
- 2. TV sets.** Add another TV by pasting the full address from its screen, including token. The television serving this page is always available as "This TV".
- 3. Media keys.** Pause and seek keys on the keyboard control the TV even when the browser is minimised. They take those keys from other programs — turn the option off if that gets in the way. A click on the page may be required if the browser blocked audio.
- 4. TMDb key.** API v3 key for descriptions and ratings. Posters work without a key.
- 5. Sources.** List, add and delete. Deleting a source also clears its list and EPG on the TV.
- 6. Backup.** Downloads a JSON file with sources, local radio playlists, favourites, history, VOD positions, manual EPG matches and program settings. The file also contains the login details stored with your sources — keep it safe. Restore replaces the current data. File limit: 10 MB.
- 7. Bouquets.** The bouquets of the selected source with a channel count each. An unticked bouquet disappears from the channel list on the TV, from bouquet switching while watching and from the rail in the Channels tab. Its channels **stay** in search, in favourites and in Films — the point is shorter lists, not hiding part of your own playlist.
- 8. TV guide.** **Fetch EPG for the selected source** starts an XMLTV import on the TV. Below it sits the **Manual EPG matching** card — described in chapter 15.
- 9. Channel and radio station list.** Fetch the playlist from the provider again (same as the blue remote key). Below the divider: hiding VOD items on the channel list and the **next-programme notice** — how many minutes ahead it should appear, or "off".

10. **Guide stored on the TV.** How many programmes for how many channels sit in the database, how much space they take and **how far ahead the guide runs** — that is, when it needs refreshing. The **Clear the guide** button removes programmes and automatic matches; **manual matches are kept** and come back into play at the next import.
11. **Storage.** The TV (channel logos and film posters, each with its own limit) and this browser (a copy of the lists so the panel opens instantly) are listed separately. Clearing browser storage deletes nothing on the TV.

15. Manual EPG matching PANEL

The program matches a channel to the guide by name. On most channels that is enough, but a handful always remain whose provider name has nothing in common with the name in the XMLTV file — such a channel shows *no EPG data* even after a correct import. From this version you can point it at the right entry by hand.

The **Manual EPG matching** card is in **Settings** → **TV guide**, below the import button. It applies to the source selected in the panel header; radio does not use it.

HPL ink Settings — Manual EPG matching

Matched 1312 of 1397 · manual 3

CHANNEL FROM THE SOURCE	CHANNEL IN XMLTV
Type part of the channel name • TVP HD → no match • TVN HD → TVN (96%) Quality 96% · automatic · TVN	Type the EPG channel name • TVP HD [tvp.hd.pl] • TVP 1 HD [tvp1.hd.pl] Save mapping Restore automatic

1. Find the channel

Type part of the name in the left field. The list shows what the channel is matched to, or *no match*.

2. Check the quality

Below the list is a percentage and the matching method: *exact*, *automatic* or *manual*.

3. Point at an entry

Find the right XMLTV channel in the right field and select **Save mapping**.

4. Fetch the EPG

Run the import again — only then do that channel's programmes reach the database.

- The card header gives the balance: how many channels of the source have a guide and how many were matched by hand.
- The right list is the catalogue of channels from the imported XMLTV file, with the identifier in square brackets. The search box above it is pre-filled with the name of the selected channel.
- **Restore automatic** removes the manual decision for the selected channel and hands it back to the automatic matching.
- Manual assignments win over the automatic ones and **survive later EPG imports**. Only deleting the source clears them.
- They also go into the backup and come back when it is restored.

When the guide appears at once and when only after an import: HD and SD channels of the same broadcaster share one set of programmes. If the XMLTV entry you pick is already used by another channel, the guide shows immediately. If not, the programmes arrive with the next import.

16. Troubleshooting

The ACTIVATION REQUIRED screen returns or the code is rejected

- Send the request code copied from **this** television.
- Use the exact subject `HPL IPTV E2+ Radio activation`.
- Paste the entire code you received, with no trailing space and nothing cut off.
- A code from another television will not work on this one.

The panel does not open

1. Check that the app is running on the TV — the panel works only then.
2. The computer and the TV must be on the same network. Turn off Wi-Fi client isolation on the router.
3. Paste the full address from the screen, including `token`.
4. If the TV shows a red message that the panel is not working, start the app again.

The source is there, but the list is empty

Open the source on the TV or use **Fetch the selected playlist again** in the panel. The panel keeps a copy of the list in the browser; adding a source does not load the channels by itself.

No EPG

- IPTV / Xstream: enter an XMLTV address on the source and run the import (green on the TV or the button in Settings).
- Enigma2: the guide should come from the receiver. Check OpenWebif and that a bouquet is loaded.
- Radio has no TV guide.

One channel has no guide while the rest do

Usually it is the name: different at the provider than in the XMLTV file, so the automatic matching has nothing to recognise the channel by. Open **Settings** → **TV guide** → **Manual EPG matching**, point at the right entry and run the import once more — chapter 15.

The picture freezes or the channel does not respond

The program retries the stream on its own. If the folder has a backup link, it switches to it. Check the provider account, the number of concurrent connections, and whether the same address plays in VLC on a computer.

The message *No picture detected. Restoring the picture* means the picture stopped while the sound kept playing and the program is rebuilding the connection — nothing to do. The message *No network · waiting for a connection* means the problem is with Wi-Fi or the router; the program returns to the picture by itself once the network is back.

The picture is stretched or has bars

Hold in the player and change the mode: **Fit** shows the whole frame with bars, **Zoom** fills the screen at the cost of the edges, **Stretch** is the default. On 16:9 material all three modes give the same picture.

A film does not start, but a TV channel does

Check that the account includes VOD and that the playlist has separate film paths. The VOD filter on the TV does not block the Films / Series tabs in the panel — it only cleans the channel list.

Film descriptions are missing

Posters do not need TMDb. Descriptions and ratings need an API v3 key in Settings and internet access in the browser.

The phone remote does not control the TV

The phone must be on the same network. Scan the code again from the panel that is currently open. Check that the remote has the correct television selected.